TECHNICAL SPECIFICATION & ENGINEERING MANUAL

SmartAttendance™ Hardware & Firmware Reference

Low-Level Subsystem Architecture, Pinout Matrix, MIPI CSI/DSI Drivers, ESP-DL Neural Inference, Power Topology, FreeRTOS Concurrency & Board Diagnostics

- **SoC & Memory:** Espressif ESP32-P4 RISC-V Dual-Core @ 400 MHz + 32MB Hex-PSRAM @ 200 MHz
- **Display Bridge:** EK79007 MIPI-DSI (2 lanes @ 900 Mbps, 51 MHz DPI clock, 1024×600 IPS panel)
- **Touch Controller:** GT911 Capacitive Touch with verified manual GPIO40/42 I2C address lock (0x5D)
- **Camera Subsystem:** SC2336 24-Pin MIPI-CSI (SCCB 0x36), hardware ISP Bayer BGGR, RAW8 720p @ 30fps
- **DMA & Concurrency:** Double-buffered frame acquisition in PSRAM (`s_detection_fb_buf`) eliminating tearing
- **Edge AI Models:** Quantized int8 ESP-DL FaceDetect (MSR/MNP) & MobileFaceNet 128D embedding extractor
- **Power & Gas Gauge:** Dual power rails (LDO3 2.5V, LDO4 3.3V), STC8 battery fuel gauge telemetry
- **Storage & DB:** SDMMC 1-bit FAT32 (`/sdcard`) with SQLite3 embedded database and backup sync

Target Audience:	Embedded Engineers, Firmware Devs, Hardware Techs	**Hardware Revision:**	CrowPanel Advanced 7″ ESP32-P4 v1.0
Framework / SDK:	ESP-IDF v5.3+ / ESP-DL v1.1.4+	**Toolchain:**	riscv32-esp-elf-gcc / FreeRTOS SMP
Document ID:	SA-ESP32P4-ENG-2026-REV4	**Publication Date:**	October 2026

[!] HARDWARE WARNING: Confidential Engineering Document

This manual contains low-level hardware schematics, register configurations, and flash partition offsets intended exclusively for technical personnel. End-user operating guidelines are documented in the companion SmartAttendance User Manual.

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Section 1: Hardware Architecture & Electrical Pinout

The Elecrow CrowPanel Advanced 7″ ESP32-P4 v1.0 integrates the ESP32-P4 SoC with external Hex-PSRAM, MIPI CSI/DSI bridges, and power management circuits. The table below represents the empirically validated hardware pin assignments.

Peripheral	GPIO Pin	Interface / Net Name	Engineering Notes
Backlight PWM	GPIO 31	LEDC PWM Channel 0	30 kHz carrier frequency, 11-bit resolution (duty 200-2047).
Touch SDA	GPIO 45	I2C_NUM_0 SDA	Requires external 4.7kΩ pull-up to 3.3V.
Touch SCL	GPIO 46	I2C_NUM_0 SCL	100 kHz standard clock speed with pull-up.
Touch Reset	GPIO 40	GT911_RST	Must be driven via manual GPIO pulse to fix 0x5D address.
Touch Interrupt	GPIO 42	GT911_INT	Hold low during RST release to latch address 0x5D.
Camera CSI Data	DSI/CSI PHY	MIPI-CSI 2-Lane	OmniVision SC2336 24-pin FPC ribbon; RAW8 format.
Camera SCCB SDA	GPIO 7	SCCB I2C SDA	Sensor address 0x36 (alternate address; NOT 0x30).
Camera SCCB SCL	GPIO 8	SCCB I2C SCL	Clock for camera sensor register read/write.
SDMMC Clock	GPIO 43	SD_CLK	SDMMC 1-bit clock line; 20 MHz to 40 MHz clock.
SDMMC Command	GPIO 44	SD_CMD	SDMMC command line with internal pull-up enabled.
SDMMC Data 0	GPIO 39	SD_D0	SDMMC data 0 bidirectional transfer.
Battery ADC	GPIO 4	ADC_CHANNEL_4	Monitors 3.3V to 4.2V battery via resistor divider network.
DS1302 RTC CLK	GPIO 51	J7 Header IO51	Serial clock for external battery-backed RTC.
DS1302 RTC DAT	GPIO 50	J7 Header IO50	Bidirectional serial data line for RTC register transfers.
DS1302 RTC CE	GPIO 49	J7 Header IO49	Chip Enable (active high) for DS1302 bus access.

[X] CRITICAL CONSTRAINT: Power Rail Sequencing (LDO3 & LDO4)

The EK79007 display bridge and MIPI PHY require internal LDO3 (2.5 V) and LDO4 (3.3 V) rails. If firmware fails to initialize these PMU regulators prior to display driver initialization, the MIPI-DSI link will fail to negotiate clock synchronization, leaving the panel permanently dark.

Section 2: MIPI-DSI Display & Backlight Subsystems

The display panel connects via a 2-lane MIPI-DSI interface managed by an EK79007 driver IC. Key physical parameters:

- **Lane Bit Rate:** 900 Mbps across 2 differential data lanes.
- **DPI Pixel Clock:** 51 MHz nominal.
- **Resolution & Porch Timing:** 1024×600 active pixels. HSYNC pulse: 20, HBP: 140, HFP: 160. VSYNC pulse: 3, VBP: 20, VFP: 12.
- **Critical Driver Constraint:** Do **NOT** call esp_lcd_panel_disp_on_off(). The EK79007 bridge does not support software display power toggle; attempting to do so locks the MIPI PHY.
- **LVGL 9 Optimization:** Allocate draw buffers strictly from PSRAM: swap_bytes=false, buff_spiram=true, sw_rotate=true, double_buffer=true.
- **Backlight Formula:** LEDC 30 kHz on GPIO31. Duty formula: duty = (pct != 0) ? ((pct * 18) + 200) : 0.

Section 3: GT911 Capacitive Touch Controller Quirks

The Goodix GT911 capacitive touch IC requires a precise hardware reset timing sequence to latch its I2C slave address at 0x5D. Standard generic drivers fail because they do not control RST and INT with sub-millisecond precision.

```
// 1. Install legacy I2C driver on I2C_NUM_0 (GPIO 45 SDA, GPIO 46 SCL, 100 kHz)
i2c_config_t i2c_conf = {
    .mode = I2C_MODE_MASTER, .sda_io_num = 45, .scl_io_num = 46,
    .sda_pullup_en = GPIO_PULLUP_ENABLE, .scl_pullup_en = GPIO_PULLUP_ENABLE,
    .master.clk_speed = 100000,
};
i2c_param_config(I2C_NUM_0, &i2c;_conf);
i2c_driver_install(I2C_NUM_0, I2C_MODE_MASTER, 0, 0, 0);

// 2. Hardware timing pulse to force address 0x5D:
gpio_set_direction(40, GPIO_MODE_OUTPUT); // RST
gpio_set_direction(42, GPIO_MODE_OUTPUT); // INT
gpio_set_level(40, 0); gpio_set_level(42, 0);  // Both driven low
vTaskDelay(pdMS_TO_TICKS(1));
gpio_set_level(40, 1);                          // Release RST high while INT stays low
vTaskDelay(pdMS_TO_TICKS(50));
gpio_set_level(42, 1);                          // Release INT high -> Latches 0x5D address!
vTaskDelay(pdMS_TO_TICKS(20));

// 3. Configure esp_lcd_panel_io_i2c (16-bit register commands are MANDATORY):
esp_lcd_panel_io_i2c_config_t io_cfg = {
    .dev_addr = 0x5D, .control_phase_bytes = 1, .dc_bit_offset = 0,
    .lcd_cmd_bits = 16,     // MUST be 16, NOT 8!
    .lcd_param_bits = 16,   // MUST be 16, NOT 8!
    .flags.disable_control_phase = 1,
};

// 4. Touch driver configuration (RST and INT set to NC because handled above):
esp_lcd_touch_config_t tp_cfg = {
    .x_max = 1024, .y_max = 600,
    .rst_gpio_num = GPIO_NUM_NC, .int_gpio_num = GPIO_NUM_NC,
    .flags = { .swap_xy = 0, .mirror_x = 1, .mirror_y = 1 },
};
esp_lcd_touch_new_i2c_gt911(tp_io, &tp;_cfg, &tp;);
vTaskDelay(pdMS_TO_TICKS(1000));
esp_lcd_touch_exit_sleep(tp); // Mandatory GT911 wake-up command
```

Section 4: MIPI-CSI Camera Driver & ISP Pipeline

The camera sensor is an OmniVision SC2336 connected to the ESP32-P4 hardware ISP via 2-lane MIPI-CSI:

- **SCCB Bus Address:** 0x36 (7-bit address; not 0x30).
- **Bayer Pixel Format:** Native sensor output is **BGGR** RAW8. Do NOT enable HMIRROR without switching ISP Bayer order to GBRG, or severe chromatic distortion will result.
- **Resolution & Framerate:** MIPI_2lane_24Minput_RAW8_1280x720_30fps (the only hardware-supported profile on this board revision).
- **Indoor Exposure Baseline:** exposure_us = 15000, exposure_val = 1100, gain = 4000.
- **Semaphore Callback Architecture:** Never call esp_cam_ctlr_receive() in a loop. Register an on_get_new_frame callback that gives a FreeRTOS semaphore; the consumer task blocks on the semaphore to retrieve frames without dropping DMA interrupts.
- **Double-Buffering DMA Protection:** A dedicated buffer s_detection_fb_buf is allocated in PSRAM during boot. When the detection task is ready, the camera task copies the frame atomically, ensuring the face detector and landmark alignment never read a partially overwritten buffer.

Section 5: High-Speed PSRAM & Flash Memory Architecture

The ESP32-P4 requires 32MB Hex-mode PSRAM running at 200 MHz to support simultaneous 720p MIPI video decoding, LVGL frame rendering, and neural network tensor allocations.

```
# Mandatory symbols in sdkconfig / sdkconfig.defaults.esp32p4:
CONFIG_IDF_EXPERIMENTAL_FEATURES=y          # Required to unlock 200MHz PSRAM
CONFIG_SPIRAM=y
CONFIG_SPIRAM_MODE_HEX=y                    # 16-line Hex-mode bus
CONFIG_SPIRAM_SPEED_200M=y                  # 200 MHz clock rate
CONFIG_SPIRAM_USE_MALLOC=y
CONFIG_SPIRAM_ALLOW_BSS_SEG_EXTERNAL_MEMORY=y
CONFIG_SPIRAM_RODATA=y
CONFIG_ESPTOOLPY_FLASHSIZE_16MB=y           # 16 MB QSPI Flash
CONFIG_I2C_SKIP_LEGACY_CONFLICT_CHECK=y    # Avoids driver/i2c legacy symbol clash
```

[X] CRITICAL CONSTRAINT: Stack Allocation Anti-Pattern

Never allocate large frame arrays on the FreeRTOS task stack (e.g., uint8_t crop[112*112*3]). Doing so causes an immediate task stack overflow. All vision tensors and image crops must be dynamically allocated from PSRAM using heap_caps_malloc(size, MALLOC_CAP_SPIRAM) and freed upon task completion.

Section 6: Power Supply, Battery Telemetry & STC8 Gauge

The power management system combines continuous USB-C DC input with single-cell LiPo battery backup:

- **ADC Sensing Network:** GPIO 4 (ADC1 Channel 4) measures battery terminal voltage through a 2:1 resistive divider. Nominal voltage range: 3.30V (empty cutoff) to 4.20V (full charge).
- **STC8 Gas Gauge Protocol:** An onboard STC8 coprocessor continuously samples cell voltage and communicates smoothed battery percentages and charging status flags to the ESP32-P4 over internal serial telemetry.
- **Threshold Gates:**
  - **15% Warning:** Flashes low battery alert banner.
  - **5% Emergency Shutdown:** Calls db_sync_to_disk(), unmounts /sdcard, and pulls display/camera power rails low to protect flash data from brownout corruption.
- **Calibration Logger:** Firmware includes a technical calibration module (`sd_logger.c`). Setting `BATTERY_CALIB_LOGGING=1` in `config.h` records raw ADC mV, STC8 reported percentage, and system state to `/sdcard/logs/batt_calib.csv` every 20 seconds.

Section 7: Deep Learning Vision Engine & Face Embeddings

The biometric engine uses Espressif ESP-DL quantized neural network models executed on the RISC-V SIMD architecture:

- **Model Locations on SD Card:**
  - /sdcard/models/p4/human_face_detect_msr_s8_v1.espdl (Fast multi-scale face proposal)
  - /sdcard/models/p4/human_face_detect_mnp_s8_v1.espdl (5-point keypoint landmark detection)
  - /sdcard/models/p4/mfn_s8_v1.espdl (MobileFaceNet 128D int8 embedding extractor)
- **Keypoint Indexing (ESP-DL vs MTCNN):** ESP-DL outputs 5 landmarks in the order: [Left Eye, Left Mouth, Nose, Right Eye, Right Mouth]. Swapping landmarks causes head yaw calculation to explode to 900°, breaking the quality filter. Indices must be mapped 1-to-1 without reordering.
- **Affine Transform:** Warps cropped face keypoints to fixed coordinate anchors in a canonical 112×112 RGB565 matrix.
- **Feature Extraction:** MobileFaceNet outputs a 128-dimensional signed int8 vector $e \in [-128, 127]^{128}$.

Section 8: Multi-Frame Temporal Consensus & Margin Gating

To achieve near-zero false acceptance rates without sacrificing throughput:

- **Sliding Window Accumulator:** Tracks up to 6 valid frames over a 1500ms window.
- **Early Exit Fast Path:** If 3 consecutive frames show unanimous candidate matching with similarity $S \geq 0.80$, verification completes immediately.
- **Dynamic Margin Gate:** Required separation margin over runner-up candidate #2 depends on Top-1 similarity:
  - **$S \geq 0.78$:** Requires $\Delta S \geq 0.03$.
  - **$0.74 \leq S < 0.78$:** Requires $\Delta S \geq 0.05$.
  - **$0.70 \leq S < 0.74$:** Requires strict $\Delta S \geq 0.08$.
- **Specular Glare Filter:** Subsampled check rejecting frames with >15% pixels saturated at luminance Y > 240.
- **Mouth Opening Gate:** Computes vertical landmark ratio (Y_mouth - Y_nose) / eye_dist. Frames with ratio > 0.85 are rejected to prevent expression deformation.

Section 9: FreeRTOS Task Architecture & Mutex Locking

The firmware executes across two symmetric RISC-V cores. Strict task priority and synchronization rules prevent race conditions:

Task Name	Priority	Stack Size	Core Affinity & Architectural Role
LVGL UI Task	10 (Highest)	8,192 bytes	Core 1. Handles screen rendering and touch event dispatch.
Detection Task	8	32,768 bytes	Core 0. Runs ESP-DL neural face detector and landmark alignment.
Camera Task	7	8,192 bytes	Core 0. Receives MIPI DMA interrupts and manages frame buffers.
Database Task	6	8,192 bytes	Core 1. Executes SQLite3 attendance writes and user queries.
Network Task	5	16,384 bytes	Core 0. Manages Wi-Fi, HTTPS sync API, and Telegram bot calls.
Battery Task	4	4,096 bytes	Core 1. Periodic 10s ADC voltage and STC8 telemetry polling.

LVGL Thread Safety Rule: Any non-display task invoking LVGL widget functions must acquire ui_acquire(). If ui_acquire() returns false (timeout), the caller MUST skip the UI call and bypass ui_release() to avoid corrupting the FreeRTOS recursive semaphore.

Section 10: Partition Table & Dual-Bank OTA Flashing

The 16 MB flash memory is partitioned for safe dual-bank A/B firmware upgrades:

```
# Name,     Type, SubType, Offset,   Size,     Flags
nvs,        data, nvs,     0x9000,   0x6000,
otadata,    data, ota,     0xf000,   0x2000,
phy_init,   data, phy,     0x11000,  0x1000,
ota_0,      app,  ota_0,   0x20000,  0x600000, # 6 MB Application Bank 0
ota_1,      app,  ota_1,   0x620000, 0x600000, # 6 MB Application Bank 1
storage,    data, spiffs,  0xc20000, 0x3d0000,
```

Flashing Commands:

• **Full Clean & Build:** idf.py fullclean && idf.py build

• **Flashing via COM port:** esptool.py --chip esp32p4 -p COM5 -b 460800 --before default_reset --after hard_reset write_flash 0x0 build/bootloader/bootloader.bin 0x10000 build/partition_table/partition-table.bin 0x20000 build/SmartAttendance.bin

Section 11: SQLite3 Schema & On-Disk Data Structures

Biometric profiles and attendance events are stored in /sdcard/attendance.db using SQLite3:

```
-- Users Table (Biometric Templates & Biographical Data)
CREATE TABLE IF NOT EXISTS users (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    uuid TEXT UNIQUE NOT NULL,
    name TEXT NOT NULL,
    student_id TEXT UNIQUE,
    phone_number TEXT,
    telegram_id TEXT,
    role TEXT NOT NULL CHECK(role IN ('student','lecturer','admin')),
    face_embedding BLOB,               -- 128-byte int8 MobileFaceNet vector
    created_at INTEGER NOT NULL,
    updated_at INTEGER,
    enroll_quality INTEGER DEFAULT 0,  -- Quality score (0 - 100)
    enroll_accepted INTEGER DEFAULT 0,
    enroll_rejected INTEGER DEFAULT 0,
    model_version INTEGER DEFAULT 1,
    embedding_dim INTEGER DEFAULT 128
);

-- Attendance Log Table
CREATE TABLE IF NOT EXISTS attendance (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    user_id INTEGER NOT NULL,
    session_id INTEGER,
    course_code TEXT,
    timestamp INTEGER NOT NULL,
    confidence REAL NOT NULL,
    synced INTEGER DEFAULT 0,          -- 0=local only, 1=synced to cloud
    FOREIGN KEY (user_id) REFERENCES users(id)
);
```

Section 12: Board-Level Diagnostics & Hardware Repair

Technicians diagnosing hardware failures should follow this empirical troubleshooting matrix:

Hardware Symptom	Diagnostic Test Point	Root Cause & Repair Procedure
Board fails to enumerate on USB	Measure 5V rail at input diode; check 3.3V LDO output.	If 3.3V rail is shorted (< 100Ω to GND), check decoupling capacitors near ESP32-P4 SoC for ESD breakdown.
Display blank; backlight off	Probe GPIO31 with oscilloscope; check LDO3 (2.5V) and LDO4 (3.3V).	Verify 30 kHz PWM carrier on GPIO31. If PWM is present but backlight is off, check FPC backlight driver boost circuit.
Touch I2C arbitration failure	Probe GPIO45 (SDA) and GPIO46 (SCL) with logic analyzer.	Verify external 4.7kΩ pull-up resistors to 3.3V. Inspect GPIO40 (RST) and GPIO42 (INT) for broken solder joints on 6-pin touch FPC.
Camera probe returns SCCB error	Verify 24-pin FPC latch connection; probe 3.3V analog rail.	Ensure gold contacts face towards PCB. Confirm sensor address is 0x36. Replace SC2336 camera module if SCCB fails to respond.
MicroSD read timeout	Check GPIO43 (CLK), GPIO44 (CMD), GPIO39 (D0).	Confirm 1-bit SDMMC pull-up resistor. Ensure card is partitioned as MBR FAT32 (GPT partitions are not supported by fatfs driver).

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